

Sri Hari Sirisipalli

AI Systems Engineer | Scalable ML & Retrieval Architectures

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Profile

AI Systems Engineer specializing in scalable retrieval systems, digital twin modeling, and signal-driven anomaly detection. Built high-throughput embedding pipelines supporting 1,000+ daily evaluations, executed large-scale experimentation across 88K+ simulation scenarios, and deployed AI services on cloud-native infrastructure. Experienced in end-to-end system design from raw data processing to production integration.

Professional Experience

Software Engineer – LLM & ML Infrastructure, Pangeon 03/2024 – Present | Contract
Lead development of scalable embedding-driven retrieval and LLM-based evaluation systems for large-scale patent similarity analysis.

- Developed and deployed high-throughput embedding pipelines integrated with LLM semantic scoring for automated prior-art ranking.
- Support 1,000+ patent similarity evaluations daily, enabling scalable semantic comparison across large patent datasets.
- Re-architected compute-intensive workloads from EC2 to AWS Lambda, reducing infrastructure costs by 30% and improving batch completion speed by 5–7x.
- Built asynchronous processing and multiprocessing frameworks to increase similarity evaluation throughput.
- Provisioned reproducible AWS infrastructure using Terraform to enable consistent cross-account deployments.

AI Systems Engineer, Independent Consulting & R&D 06/2023 – Present

Digital Twin Systems – Offshore Riser Modeling:

- Developed surrogate ML models trained on 88,560 simulated sea states for offshore motion and fatigue prediction.
- Executed 3,525 ML/DL experiments and ~1.08M model fits to identify optimal architecture and feature configurations.
- Processed 3,888 time-series simulation files (4 Hz sampling) to construct structured environmental prediction pipelines.
- Reduced worst-case angular prediction error by 41% through structured feature optimization and circular regression modeling.
- Established reproducible training and inference workflows with strict dataset partition control to eliminate leakage.

Engineering Simulation Automation:

- Built ANSYS APDL and MOSES batch execution pipelines to automate large-scale simulation data generation.
- Orchestrated structured extraction and transformation of simulation outputs into ML-ready datasets.
- Standardized execution workflows to improve experiment repeatability and reduce manual intervention.

Anomaly Detection & Signal Processing:

- Developed unsupervised vibration-based fault detection pipeline using 16+ time-domain and 11 frequency-domain features.
- Processed multi-day sensor time-series datasets to construct structured feature matrices for anomaly modeling.
- Applied PCA-based dimensionality reduction to enhance cluster separability prior to classification.
- Implemented clustering-based anomaly detection trained on normal operating data, achieving zero false positives during baseline validation.
- Quantified anomaly robustness using cluster compactness metrics (WCSS: 441.78) and distance thresholds.

Retrieval & Computer Vision Systems:

- Architected multi-stage similarity pipeline across 100,000+ unique cattle images.
- Indexed 100K+ embeddings using FAISS multi-index vector structures and ranking fusion strategies.
- Distributed feature computation across 20 worker processes to support high-volume similarity evaluation.

Retrieval-Augmented Generation (RAG):

- Implemented retrieval-augmented generation pipelines over multi-year internal technical report corpora.
- Designed embedding ingestion workflows including chunking, metadata filtering, and indexing strategies.
- Integrated FAISS-based similarity search into FastAPI backend services.
- Reduced transcription latency by 75% and synthesis time by 52% through backend optimization.

Data Engineer, Sas2Py 03/2023 – 06/2023 | Full-Time

- Led migration of legacy SAS data pipelines to PySpark, redesigning batch workflows for distributed execution and improved scalability.
- Built automated validation framework ensuring >99% functional parity between SAS and PySpark outputs across production workloads.
- Modeled job dependency graphs to eliminate scheduling bottlenecks and optimize distributed execution efficiency.

Machine Learning Intern, Corteva Agriscience 07/2022 – 12/2022 | Internship

- Converted TensorFlow and PyTorch models to ONNX for cross-platform deployment compatibility.
- Optimized inference performance using structured pruning and post-training quantization.
- Validated numerical consistency across model formats to ensure reliable production deployment.

Software Engineer Intern, Dojima Networks 06/2022 – 12/2022 | Internship

- Integrated Polkadot ecosystem components to enable cross-chain interoperability within Dojima’s blockchain infrastructure.
- Implemented cross-chain communication workflows for transaction routing and state synchronization.
- Deployed Prometheus–Grafana monitoring stack to provide real-time API observability and system health tracking.

Core Skills

Machine Learning & AI: LLMs, RAG, Deep Learning, Feature Engineering, Model Optimization, Anomaly Detection, NLP

Retrieval & Vector Systems: FAISS, Embedding Pipelines, Vector Search Architectures, Similarity Search

Cloud & Infrastructure: AWS (EC2, Lambda, SQS, IAM), Terraform, Docker, Distributed Systems, CI/CD

Data & Backend Engineering: PySpark, ETL Pipelines, FastAPI, SQL

Programming Languages: Python

Education

Bachelor of Technology (B.Tech) – Mechanical Engineering, 08/2018 – 06/2022 | Hyderabad
Mahindra University, Hyderabad

- CGPA: 7.5
- Focused on computational modeling, numerical methods, and systems analysis, forming the foundation for applied AI and large-scale modeling systems.